

CY-223 Network Security

Instructor - Adnan Bashir



My Affiliations

Islamabad, PAK

Research Faculty



2026

My Affiliations

Albuquerque, USA

Ph.D Unsupervised ML^{*}



2026

Islamabad, PAK

Research Faculty



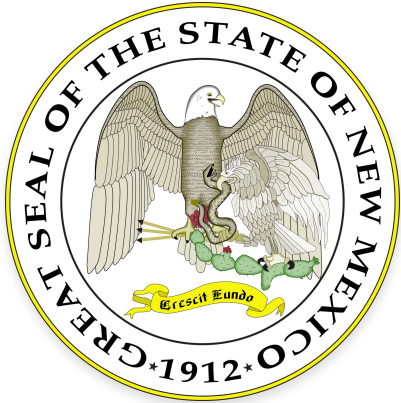
2026



My Affiliations

US State Govt.

Lead Data Steward



2025

Albuquerque, USA

Ph.D Unsupervised ML*



2026

Islamabad, PAK

Research Faculty



2026



My Affiliations

Albuquerque, USA

Music Producer (EDM)

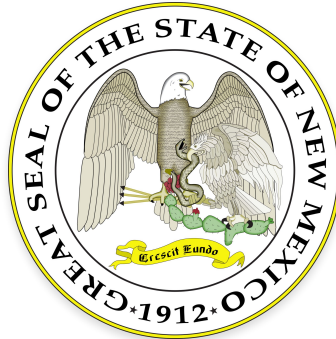
DUNAWAY PRODUCTIONS



2023

US State Govt.

Lead Data Steward



2025

Albuquerque, USA

Ph.D Unsupervised ML*



2026

Islamabad, PAK

Research Faculty



2026



My Affiliations

Albuquerque, USA

ML Engineer



2018

Albuquerque, USA

Music Producer (EDM)

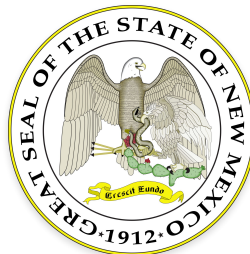
DUNAWAY PRODUCTIONS



2023

US State Govt.

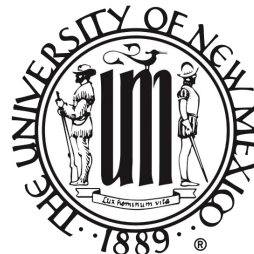
Lead Data Steward



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Ph.D Unsupervised ML*



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Islamabad, PAK

Research Faculty



2026

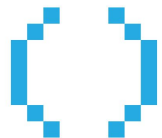
My Affiliations

Boston, MA, USA

OTF Fellow



University of
Massachusetts
Amherst



OPEN
TECHNOLOGY
FUND

2017

Albuquerque, USA

ML Engineer



2018

Albuquerque, USA

Music Producer (EDM)

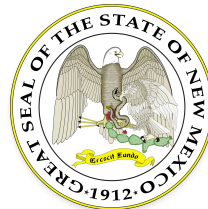
DUNAWAY PRODUCTIONS



2023

US State Govt.

Lead Data Steward



2025

Albuquerque, USA

Ph.D Unsupervised ML*



2026

Islamabad, PAK

Research Faculty



2026



My Affiliations

Lahore, PAK

Malware Analyst



2014

Albuquerque, USA	Boston, MA, USA	Albuquerque, USA	Albuquerque, USA	US State Govt.	Albuquerque, USA	Islamabad, PAK
MS(CS) ML/Data 	OTF Fellow  University of Massachusetts Amherst 	ML Engineer 	Music Producer (EDM) DUNAWAY PRODUCTIONS 	Lead Data Steward 	Ph.D Unsupervised ML ⁺ 	Research Faculty 
2016	2017	2018	2023	2025	2026	2026



My Affiliations

SEECs, Islamabad

BE (Electronics Engg.)



2012

Lahore, PAK	Albuquerque, USA	Boston, MA, USA	Albuquerque, USA	Albuquerque, USA	US State Govt.	Albuquerque, USA	Islamabad, PAK
Malware Analyst 	MS(CS) ML/Data 	OTF Fellow  University of Massachusetts Amherst 	ML Engineer 	Music Producer (EDM) DUNAWAY PRODUCTIONS 	Lead Data Steward 	Ph.D Unsupervised ML [*] 	Research Faculty 
2014	2016	2017	2018	2023	2025	2026	2026

My Affiliations



SEECs, Islamabad	Lahore, PAK	Albuquerque, USA	Boston, MA, USA	Albuquerque, USA	Dunaway Productions	US State Govt.	Albuquerque, USA	Islamabad, PAK
BE (Electronics Engg.)	Malware Analyst	MS(CS) ML/Data	OTF Fellow	ML Engineer	Music Producer (EDM) DUNAWAY PRODUCTIONS	Lead Data Steward	Ph.D Unsupervised ML*	Research Faculty
	 		 					
2012	2014	2016	2017	2018	2023	2025	2026	2026

My Affiliations



Junior



2026

SEecs, Islamabad	Lahore, PAK	Albuquerque, USA	Boston, MA, USA	Albuquerque, USA	Albuquerque, USA	State Govt., NM, USA	Albuquerque, USA	Islamabad, PAK
BE (Electronics Engg.)	Malware Analyst	MS(CS) ML/Data	OTF Fellow	ML Engineer	Music Producer (EDM)	Lead Data Steward	Ph.D Unsupervised ML*	Research Faculty
								
2012	2014	2016	2017	2018	2023	2025	2026	2026

My Affiliations

Junior

Approach
Down
Top
A
Networking
Computer



SSOR - ESORUK

2010

SEecs, Islamabad	Lahore, PAK	Albuquerque, USA	Boston, MA, USA	Albuquerque, USA	Albuquerque, USA	State Govt., NM, USA	Albuquerque, USA	Islamabad, PAK
BE (Electronics Engg.)	Malware Analyst	MS(CS) ML/Data	OTF Fellow	ML Engineer	Music Producer (EDM) DUNAWAY PRODUCTIONS	Lead Data Steward	Ph.D Unsupervised ML*	Research Faculty
								
2012	2014	2016	2017	2018	2023	2025	2026	2026

I am 88% Student

H-12, Islamabad	SEECs, Islamabad	Lahore, PAK	Albuquerque, USA	Boston, MA, USA	Albuquerque, USA	Albuquerque, USA	State Govt. , NM, USA	Albuquerque, USA	Islamabad, PAK
Approach Top Down A Networking Computer  SSOR - ESORUK	BE (Electronics Engg.) 	Malware Analyst 	MS(CS) ML/Data 	OTF Fellow  University of Massachusetts Amherst 	ML Engineer 	Music Producer (EDM) DUNAWAY PRODUCTIONS 	Lead Data Steward 	Ph.D Unsupervised ML* 	Research Faculty 
2010	2012	2014	2016	2017	2018	2023	2025	2026	2026

If I die at 65, I will be 95% Student

My Research Profile



- ⇒ Web Attack Vector Mining
- ⇒ Security Anomaly Analytics
- ⇒ Risk Modeling for Threats
- ⇒ Bot Detection Analytics
- ⇒ Malware Detection Modeling

CYBER SECURITY

7 YRS



- ⇒ Unsupervised Learning
- ⇒ LLM Based State Detection
- ⇒ Representation Learning
- ⇒ Explainable AI for Streams
- ⇒ Deep Learning for Anomalies

MACHINE LEARNING

10 YRS



- ⇒ Real Time Stream Analytics
- ⇒ Regime Shift Detection
- ⇒ Change Point Detection
- ⇒ Time Series Explainability
- ⇒ Data Access Orchestration

DATA ANALYTICS

12 YRS

My Recent Research Work

Adaptive and Unsupervised Detection of Regime Shifts and States in Data Streams

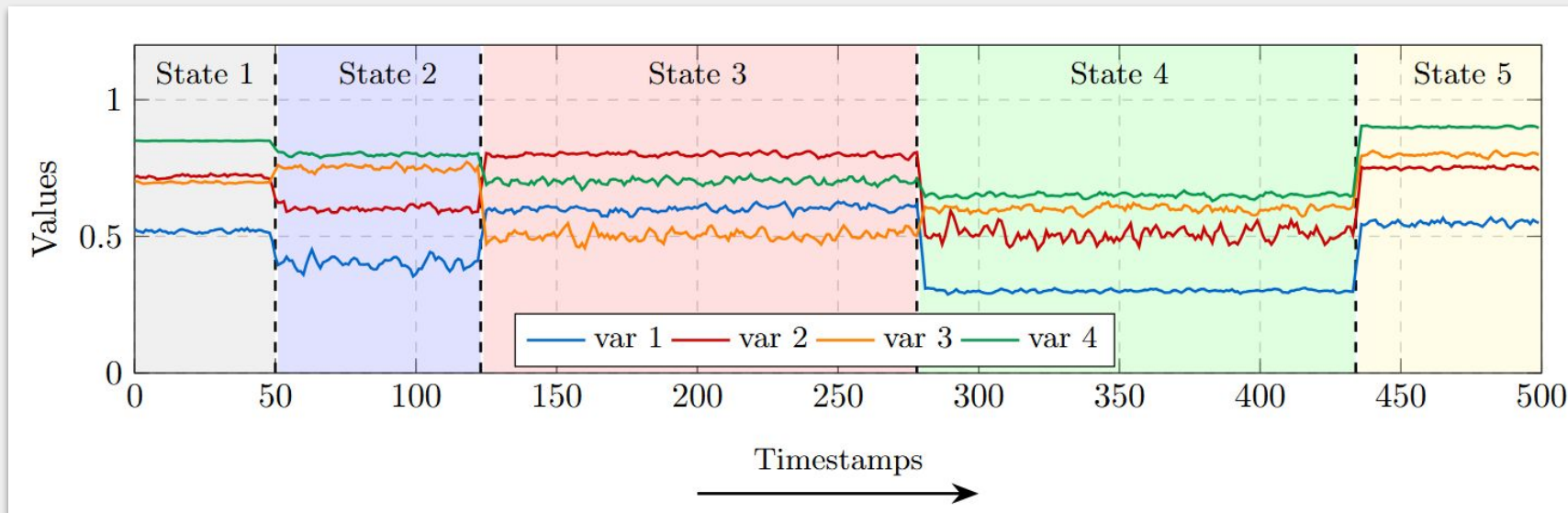


Figure: Multivariate data Stream with distinct Latent Regimes

Class Intro

What is Network Security ?

- ✧ Protect **data in transit** and **network infrastructure**
- ✧ Stop **communication attacks** (eavesdropping, spoofing, tampering, DoS etc.)
- ✧ Enforce **access control**
- ✧ **CIA (Confidentiality, Integrity, Availability)**
- ✧ **Authenticity** and **Accountability**
- ✧ **Computer security: host-only** protection
- ✧ **Network Security: traffic** and **shared network** protection

Why we need Network Security ?

- ✧ Stop **unauthorized disclosure** → **Confidentiality**
- ✧ Stop **unauthorized modification/destruction** → **Integrity**
- ✧ Stop **disruption of access/use** → **Availability**
- ✧ Security is noticed when **it fails**

Who needs Network Security ?

- ✧ Any organization handling **digital information** daily
- ✧ Anyone sending **data** over **shared networks**
- ✧ **Privacy** heavy **domains: health, finance, education**
- ✧ Proprietary data: **business, research, IP**
- ✧ **Admins** need policies, controls, monitoring
- ✧ **Users** need safe access and awareness

Do we need Computer Networks Background ?

- ✧ Helpful for understanding **TCP/IP** layers quickly
- ✧ Know basics: **IP, ports, routing, DNS**
- ✧ Deep OSI protocol detail ?
- ✧ We focus on concepts, not packet wizardry
- ✧ You will learn essentials as we go

CY-223 Learning Objectives

- ✧ Define **security attacks**, **services**, and technical **mechanisms**.
- ✧ Distinguish between **threats**, **vulnerabilities**, and **active attacks**.
- ✧ Categorize attacks into primary **passive** and **active** classes.
- ✧ Analyze **message release** and **stealthy traffic analysis** methods.
- ✧ Map protective security **services** to specific **defense** mechanisms.
- ✧ Evaluate theoretical security concepts using real world **case studies**.

Remember Remember - 5th of November

- ☞ **Cheating** results in **course failure**.
- ☞ **Plagiarism** results in **course failure**.
- ☞ If **LLMs** are used, share **prompts** and **chats** with the class (**No Exceptions**).
- ☞ Missed quiz/lab/assignment? Better have a **genuine excuse**.
- ☞ **No phones** allowed unless I give you a task.
- ☞ Need to chat? Leave and come back when done.
- ☞ **DO NOT** disturb the class whatsoever.

Course Outline

Week	Topic Description
1	Introduction, Security Attacks, Services, Mechanisms
2	X.800 Security Architecture, Layered Defense, Zero Trust
3	Enterprise Networks, Firewalls, IDS, IPS
4	DMZ Design, Traffic Flow, Data Loss Prevention
5	Network Scanning and Reconnaissance
6	Web Security: HTTP, HTTPS, Cookies, Sessions, CSRF
7	TLS, SSL, Certificates, PKI
8	VPN Architecture and Tunneling Protocols

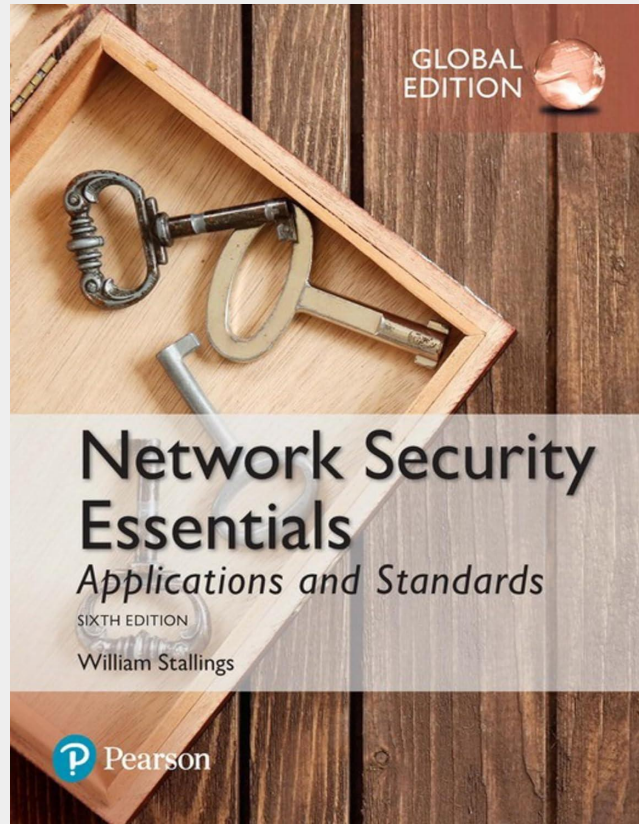
Course Outline

Week	Topic Description
9	IPsec Architecture, AH, ESP, Modes
10	IPsec Key Management, IKE, ISAKMP
11	Firewall Architectures and Management
12	IDS, IPS, Unified Threat Management
13	SOC Architecture and Incident Response
14	Interview Methods and Techniques
15	Revision
16	Project Presentations

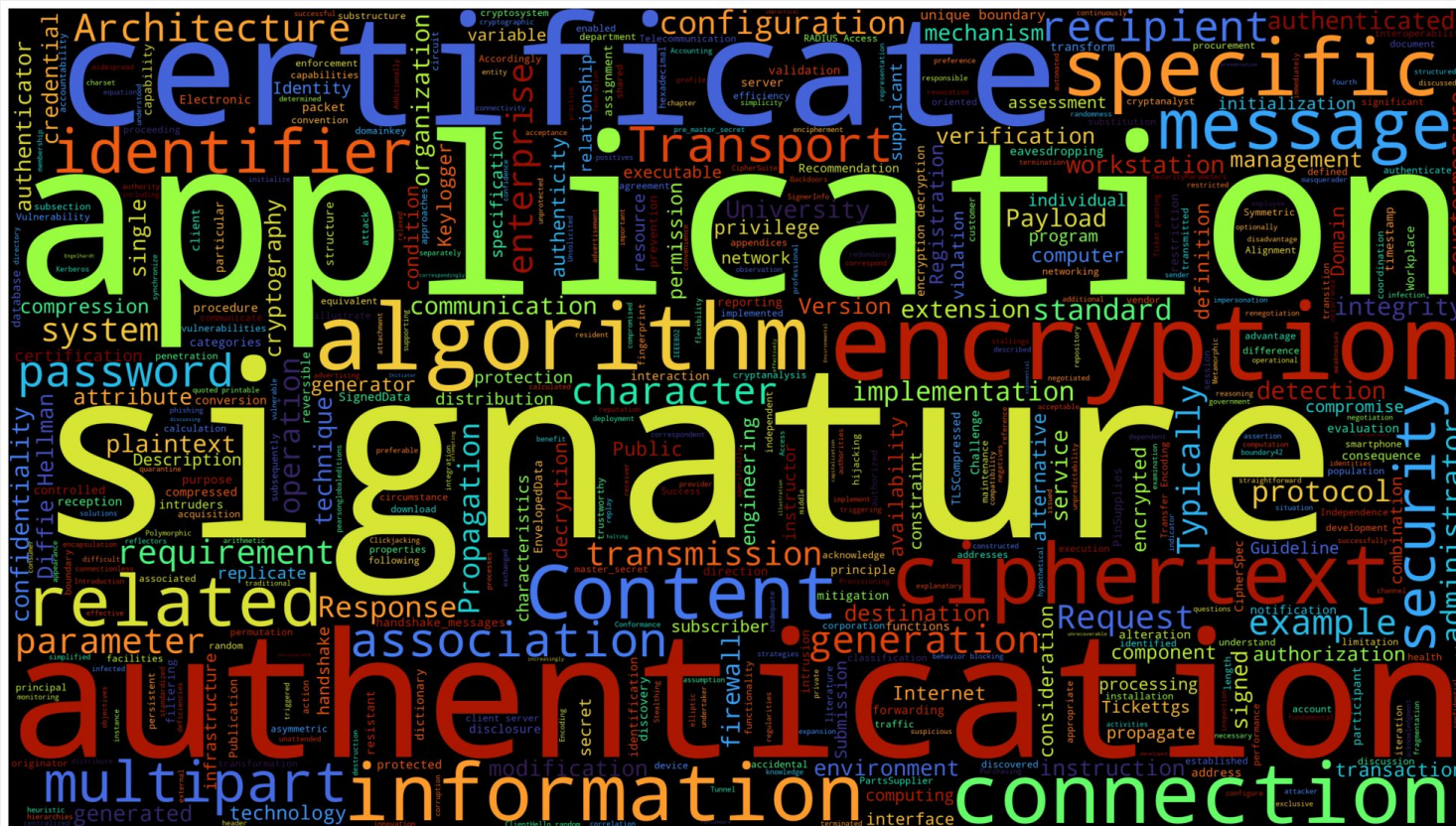
Course Breakdown

ITEM	#	%
Assignments	5	5%
Quizzes	5	5%
Midterm	1	25%
Final Exam	1	45%
Class Project	1	20%

Network Security Essentials: Applications and Standards



T1: Network Security Essentials: Applications and Standards



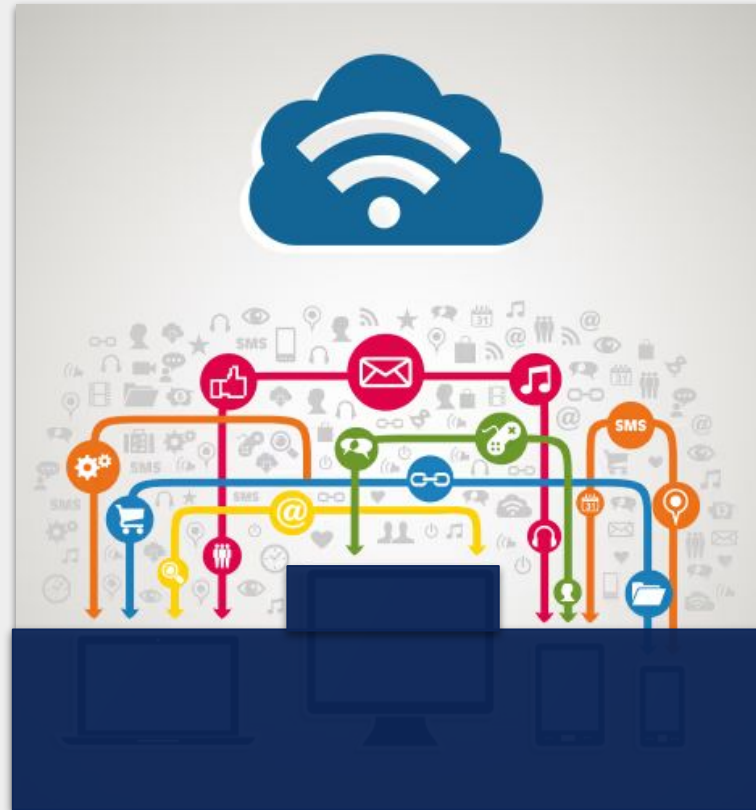
Introduction to Network Security

Computer Security

Network Security



Introduction to Network Security



Network Security

- The connection and traffic between devices.
- Protecting transmitted data (Data in Transit).
- Firewalls, VPNs, IDS/IPS.

Introduction to Network Security

Computer Security

- Hardware/Software on a specific device.
- Protecting stored data (Data at Rest).
- Antivirus, Passwords, Encryption.



Introduction to Network Security

Computer Security

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- Protecting stored data (Data at Rest).
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Network Security

- The connection and traffic between devices.
- Protecting transmitted data (Data in Transit).
- Firewalls, VPNs, IDS/IPS.

Introduction to Network Security

Displacement	Hex codes
0000(0000)	FA E9 4A 01 34 12 00 07 14 00 01 00 00 00 00 20
0016(0010)	20 20 20 20 20 20 57 65 6C 63 6F 6D 65 20 74 6F
0032(0020)	20 74 68 65 20 44 75 6E 67 65 6F 6E 20 20 20 20
0048(0030)	20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
0064(0040)	20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
0080(0050)	20 28 63 29 20 31 39 38 36 20 42 61 73 69 74 20
0096(0060)	26 20 41 6D 6A 61 64 20 28 70 76 74 29 20 4C 74
0112(0070)	64 2E 20 20 20 20 20 20 20 20 20 20 20 20 20 20
0128(0080)	20 42 52 41 49 4E 20 43 4F 4D 50 55 54 45 52 20
0144(0090)	53 45 52 56 49 43 45 53 2E 2E 37 33 30 20 4E 49
0160(00A8)	5A 41 4D 20 42 4C 4F 43 4B 20 41 4C 4C 41 4D 41
0176(00B0)	20 49 51 42 41 4C 20 54 4F 57 4E 20 20 20 20 20
0192(00C0)	20 20 20 20 20 20 20 20 20 20 20 4C 41 48 4F 52
0208(00D0)	45 2D 50 41 4B 49 53 54 41 4E 2E 2E 50 48 4F 4E
0224(00E0)	45 20 3A 34 33 30 37 39 31 2C 34 34 33 32 34 3B
0240(00F0)	2C 32 38 30 35 33 30 2E 20 20 20 20 20 20 20 20

ASCII value
-0J04↑●Π0
Welcome to
the Dungeon

(c) 1986 Basit
& Amjad (put) Lt
d.
BRAIN COMPUTER
SERVICES.. 730 NI
ZAM BLOCK ALLAMA
.IQBAL TOWN
LAHDR
E-PAKISTAN..PHJN
E :430791,443248
,280530.

Quiz 1 - Categorize the following scenario as Computer Security, Network Security, or Neither:

Scenario 0: A company installs a **Firewall** to block unauthorized traffic from entering the office LAN.

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Scenario 0: A company installs a **Firewall** to block unauthorized traffic from entering the office LAN.

Correct Categorization: Network Security

Reasoning: Firewalls control the flow of data across interconnected networks.

Quiz 1 - Categorize the following scenario as Computer Security or Network Security or Neither

Scenario 1: A student uses a brute-force script to try every possible combination to log into a professor's PC.

..... /

Scenario 2: A hacker intercepts a file containing payroll records while it is being uploaded to a cloud server.

..... /

Scenario 3: A security guard at the gate of a military base checks the national identity cards (CNICs) of all visitors before allowing them to enter the facility.

.....

Scenario 4: A user encrypts a folder on their hard drive so that no one else can read their personal tax documents.

..... /

Scenario 5: A bank manager uses a filing cabinet with a heavy combination lock to store sensitive loan documents.

.....

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Computer Security

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Network Security

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Neither

Scenario 4: A user encrypts a folder on their hard drive so that no one else can read their personal tax documents.

Computer Security

Scenario 5: A bank manager uses a filing cabinet with a heavy combination lock to store sensitive loan documents.

Neither

Allah Hafiz .